

Virtual Reality Point Cloud Annotation

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Figure 1: LIDAR scanned cotton plants visualized as point clouds in various states of annotation used for the user study. The two young cotton plants on the left were used for an annotation task, and the two mature plants on the right were used for a cotton boll counting task.

ABSTRACT

This work presents an immersive headset-based virtual reality visualization and annotation system for point clouds, oriented towards application on laser scans of plants. The system can be used to paint regions or individual points with fine detail, even with large, dense point clouds. A non-immersive desktop interface was designed for comparison within the same application. A within-subjects user study (N=16) was conducted to compare these interfaces for annotation and counting tasks. Results showed a strong preference for the immersive virtual reality interface, likely as a result of perceived and actual significant differences in task performance.

Index Terms:

Human-centered computing—Virtual Reality—Computing methodologies—Virtual Reality Computing methodologies—Point-based models—

1 INTRODUCTION

Many important real-world applications obtain and process digital representations of built and natural environments. Commonly, scans of the environment are created using methods such as light detection and ranging (LIDAR) or photogrammetry, which output discrete spatial samples, known as point clouds. Tasks involving human interaction with these point clouds, such as counting, measuring, and annotation, may be well-suited for immersive virtual reality (VR), which, through stereoscopic, head-tracked rendering, could provide a better sense of measurement depth and also improve navigation performance. Further, tracked controllers commonly used in VR may improve selection of point data. Though immersive LIDAR visualization tools exist (e.g. LIDAR Viewer [1]), there is little evidence that they improve task performance over desktop-based tools (e.g. Cloud Compare or Mesh Lab). To both address and better understand this gap, we aimed to design a VR-based tool. This paper reports on that tool, including a study that addresses the critical question of how VR may improve task performance relative to desktop tools.

Our work builds upon existing efforts [2–4], with a focus on agricultural LIDAR datasets that have vast differences in point density,

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Figure 2: Screenshot of the counting task in the VR condition. The tablet interface is visible, on which participants entered their subtotals when counting groups of bolls.

owing to the need to capture fine features of plants by capturing data from multiple perspectives. We focus on in-memory visualization, as our tasks are isolated to single plants at a time, and hence levels of detail offer few advantages.

Surprisingly, though many interfaces have been designed and developed to address the supposed shortcomings of desktop interfaces, few works have evaluated differences in task performance between different user interfaces. As such, alongside our designs for the immersive and non-immersive interfaces, we also provide evidence of how the immersive interface improves certain aspects of the annotation process.

2 SYSTEM DESIGN

The raw data used for our system were obtained using multiple scans from a Faro Scanner Focus 3D (see Figure 1). The data were then exported into the Point Cloud Library's point cloud data (PCD) file format. From here, they can be processed by our application¹ which was designed entirely within the Unity3D game engine (version 2020.3 LTS). Unity does not natively load PCD files, which required a custom importer and exporter to be built.

In order to maintain high visual quality and detail while remaining performant with the high requirements of a VR-enabled application,

¹Source is available at [blinded for review].

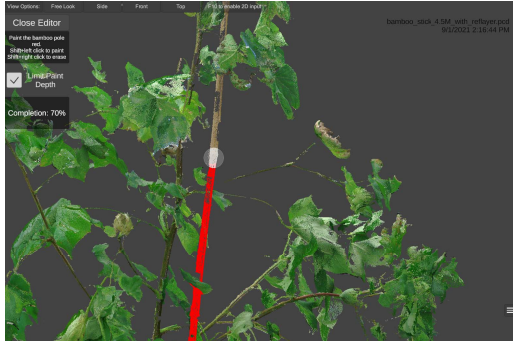


Figure 3: Screenshot of the annotation task in the 2D condition. The semi-transparent circular cursor can be adjusted in size.

our system implemented a continuous LOD similar to that presented in the work by Schütz et al. [3]. However instead of changing density of the point cloud based on view direction and head gaze, our system uses the hand positions to smoothly increase the point cloud render density in the target area. This allowed the area of focus to be guaranteed to have the maximum density, while limiting the total number of points to a modest point budget. To achieve this, a compute shader is dispatched on the GPU every frame to select the appropriate subset of points to render using a geometry shader, ranging from a circular shape to a quad, depending on distance from the camera, based on Keijiro Takahashi's PCX library (<https://github.com/keijiro/PCx>).

Beyond visualization, performance was a major concern when developing realtime annotation, and again compute shaders were used heavily to speed up the process. For every frame that annotation is indicated over a designated region, a compute shader performs a distance calculation on the GPU for every point. If the point is within the region, the byte corresponding to the current layer index is changed to match the desired layer. During the rendering process, this layer index is used by the fragment shader to draw the points in either original color, a distinct solid color, or to hide the point. This allows the operation to proceed in parallel entirely on the GPU, greatly accelerating the process.

2.1 User Interfaces

Two interfaces were designed for the study, which we abbreviate as "VR" and "2D" going forward. The VR interface was designed for use with a stereoscopic VR headset and two 6-DoF controllers. The 2D interface was designed for a single-monitor, keyboard and mouse setup. These interfaces could be swapped in real time via a keyboard switch or by clicking on a button.

The VR condition interface is depicted in Figure 2. A virtual tablet was used to show task-relevant UI elements, which was most prominent in the counting task. In order to assign points to a particular label, a painting metaphor was used. In the VR condition, this was done by visualizing a sphere around each of the users' hands, and allowing them to mark all the points within the sphere while holding a button. A secondary controller button was used to erase previously painted points and return them to the original color. The spheres could be scaled by moving the thumbstick on its Y axis to allow for more broad or precise painting actions.

The 2D condition interface is shown in Figure 3. The on-screen interface, though custom designed, mimics the features found in other annotation systems, including the ability to choose from a variety of aligned orthographic views of the cloud, or to "free look" with the mouse. The selection of a specific sphere of influence is more difficult using a mouse. Initially, the system was designed to paint all points overlayed by the 2D circular cursor, essentially creating an infinitely long cylinder of influence. While this difference between

the 2D and VR conditions could be considered fair because it arose as a result of a fundamental difference between the two systems, an screen-depth limit feature was added to the 2D condition to achieve a more similar spherical influence effect, which prevented background points from being selected accidentally.

3 STUDY DESIGN

Our study (N=16) was designed to evaluate and compare the two interfaces for two tasks that are commonly performed on point clouds: precise annotation and feature counting. A within-subjects, crossover design was used, such that each participant used each interface once for each type of task, resulting in a sequence of four tasks. In the **annotation task**, participants were asked to mark the points belonging to the stem of a plant, with live completion percentages reported by comparison to a precise annotation generated earlier. In the **counting task**, participants were asked to count cotton bolls that were scattered throughout the plant. Participants were given an interface to serve as a counter, and if they desired, to "commit" that count, which would erase the currently painted areas akin to picking the cotton. The task was completed when the participant indicated that they had counted all bolls.

4 RESULTS AND DISCUSSION

Three relative preference questions were asked for each of the two tasks, including perceived speed, accuracy, and overall preference. All three questions showed significant average preference for the VR condition. The main objective evaluation metric for both the annotation and counting tasks was completion time. In the annotation task, participants were asked to continue the annotation until they reached 98% completion, but intermediate times to 80% and 90% were evaluated as well. In the annotation task using a paired t-test, the VR condition was found to be significantly faster to 80% ($p=.002$) and 90% ($p=.001$) completion, but no significant differences were found at the 98% completion threshold. For the counting task, times were found to be 16% faster on average in the VR condition, which was found to be significant ($p=.022$) using a paired t-test.

5 CONCLUSION

This work designed and evaluated the potential of an immersive VR system for point cloud visualization and annotation of agricultural LIDAR datasets. The system enabled users to annotate, at the individual point level, large clouds at interactive frame rates in immersive VR with the help of a continuous LOD system that emphasized details at controller locations. We also designed a desktop interface for non-immersed use and comparison in a user study. Our study findings suggest that the immersive VR interface would increase both user performance and user satisfaction, though its efficacy likely diminished during latter stages of completion, requiring more detailed search and manipulation.

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